

# AI와 딥러닝을 활용한 경영분석

Week 2

# Topics

- 경영 데이터의 이해
- 고객 데이터 분석 실습
- 도서관 데이터 분석 실습

# Business Data 경영 데이터

- Any information related to a company, its operations, its customers, and its market environment. 회사, 운영, 고객, 시장 환경과 관련된 모든 정보
  - Customer data: Info from buyers, like names or purchases. 구매자들로부터 받은 정보, 예를 들어 이름이나 구매 정보
  - Product Data: Stats on sales and making items. 판매와 아이템 제작에 관한 통계
  - Financial Data: Records of money coming in and going out. 돈이 들어오고 나가는 기록들
  - Marketing Data: Details on ads and web traffic. 광고와 웹 트래픽에 관한 세부 정보

# Structured Data vs. Unstructured Data

## 정형 데이터 (Structured Data)

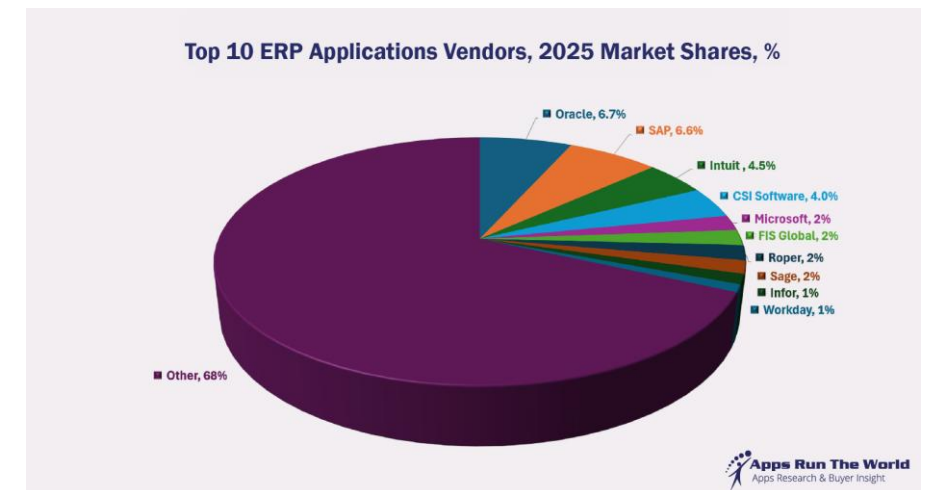
- 일정한 형식과 구조를 갖춘 데이터
- 예: 데이터베이스의 테이블, 엑셀 데이터, 고객 정보

## 비정형 데이터 (Unstructured Data)

- 정해진 구조나 형식이 없는 데이터
- 예: 텍스트 문서, 이미지, 동영상, 음성, SNS 게시물

# ERP (Enterprise Resource Planning) 기업자원관리

- Integrates all the core processes to needed to run a company into a single system. 회사를 하나의 시스템으로 운영하기 위해 핵심되는 프로세스들을 통합
- Usually include applications for accounting, human resources, sales CRM, and supply chain management. 회계, 인사, 세일, 공급 체인 관리 등을 위한 어플리케이션이 있음



# CRM (Customer Relationship Management) 고객관계관리



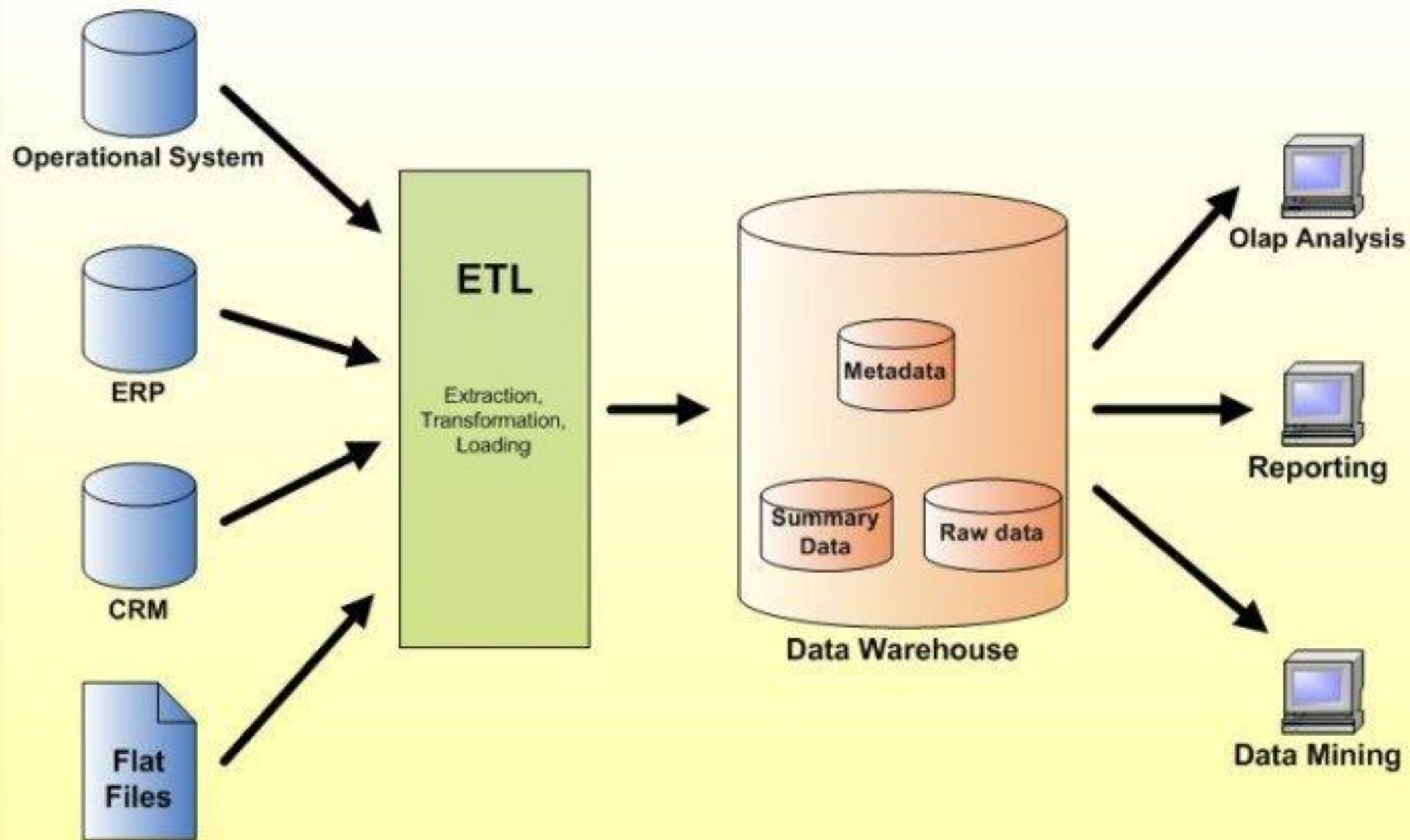
- A technology for managing all your company's relationships and interactions with customers and potential customers. 고객이나 잠재고객과의 관계를 관리하기 위한 기술들
- CRM software manages customer data, customer interaction, access business information, automate sales. 고객의 데이터, 상호작용 관리, 비즈니스 정보 접근, 세일 자동화

# Data Sources 데이터 소스들

| 데이터 소스                  | 설명                                           | 추가 구성 요소                                  |
|-------------------------|----------------------------------------------|-------------------------------------------|
| Enterprise Databases    | SQL Server, Oracle, IBM Db2 등 기업용 관계형 데이터베이스 | DB별 Client / Driver / Provider 필요할 수 있음   |
| Text Files (Flat Files) | CSV, TXT 등 텍스트 파일                            | 추가 파일 불필요                                 |
| Microsoft Excel         | .xlsx 등 Excel 파일                             | Excel 연결을 위한 ACE/OLE DB Provider 필요할 수 있음 |
| Microsoft Access        | .accdb, .mdb 등 Access 파일                     | ACE Provider 필요할 수 있음                     |
| Azure Data Sources      | Azure Blob Storage, Azure SQL Database 등     | 데이터 원본에 따라 적절한 연결 관리자/드라이버 필요             |
| Open-source Databases   | PostgreSQL, MySQL 등                          | 해당 DB의 Driver / Provider 필요               |
| ODBC Data Sources       | ODBC 드라이버를 제공하는 데이터 원본                       | ODBC Driver 필요                            |
| OLE DB Data Sources     | OLE DB Provider를 제공하는 데이터 원본                 | OLE DB Provider 필요                        |
| ADO.NET Data Sources    | .NET 데이터 공급자를 통해 연결 가능한 데이터 원본               | .NET Data Provider 필요                     |

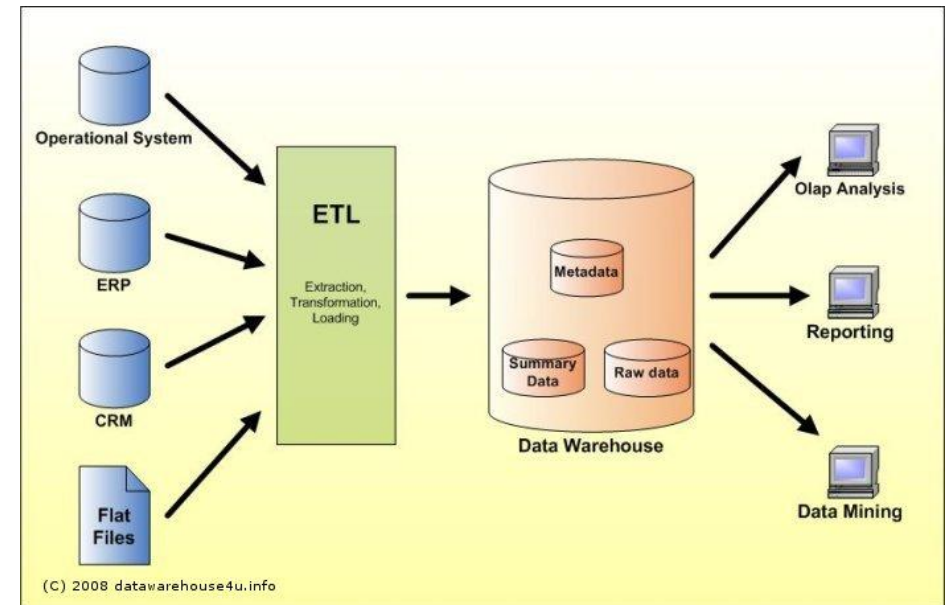
# Extraction, Transformation, and Loading (ETL)

- ETL is a three-step process. 3단계 프로세스
- Extract needed information from its source. 소스로부터 필요한 정보를 추출
- Transform the data into a standardized format. 표준화된 형태로 데이터를 변환
- Load the transformed data into a data warehouse. 변환된 데이터를 데이터창고에 적재



# Data Warehouse 데이터 창고

- A relational database that is designed for query and analysis. 질의와 분석을 위해 설계된 관계형 데이터베이스
- Usually contains historical data derived from transaction data and includes data from other sources. 거래 데이터에서 온 역사데이터나 다른 데이터 소스에서 온 데이터들을 포함



# Lab #1 고객 데이터 분석 실습

- Import XLMD\_Customer.xls and create queries to answer the following questions:  
xlmd\_customer 파일을 임포트한 후에 다음 질문에 답하시오
- Q1. Open Show all information for only customers who live in the North region. 북쪽에 사는 고객들의 정보를 보여주시오

|   | CUST ID | REGION | RENT VS. OWN | NUM HOUSEHOLD | ANNUAL INCOME | TOTAL PURCHASES | NUM PURCHASES |
|---|---------|--------|--------------|---------------|---------------|-----------------|---------------|
| 0 | 1       | North  | Own          | 1             | 10K-25K       | 38              | 1             |
| 1 | 2       | South  | Own          | 4             | 100K+         | 30              | 4             |
| 2 | 3       | North  | Rent         | 3             | 25K-50K       | 19              | 3             |
| 3 | 4       | West   | Own          | 5             | 25K-50K       | 21              | 6             |
| 4 | 5       | East   | Own          | 6             | 50K-100K      | 35              | 7             |



| Q1 | CUST ID | REGION | RENT VS# | NUM HOU | ANNUAL IN | TOTAL PUR | NUM PURC |
|----|---------|--------|----------|---------|-----------|-----------|----------|
|    | 1       | North  | Own      | 1       | 10K-25K   | \$38      | 1        |
|    | 3       | North  | Rent     | 3       | 25K-50K   | \$19      | 3        |
|    | 9       | North  | Own      | 5       | 50K-100K  | \$30      | 8        |
|    | 11      | North  | Own      | 1       | 50K-100K  | \$21      | 4        |
|    |         | North  | Own      | 6       | 50K-100K  | \$129     | 5        |
|    | 20      | North  | Rent     | 1       | 50K-100K  | \$119     | 3        |
|    | 24      | North  | Rent     | 6       | 10K-25K   | \$129     | 7        |
|    | 28      | North  | Own      | 4       | 50K-100K  | \$115     | 8        |
|    | 34      | North  | Own      | 6       | 100K+     | \$45      | 3        |
|    | 35      | North  | Own      | 5       | 10K-25K   | \$29      | 1        |
|    | 41      | North  | Own      | 7       | 25K-50K   | \$19      | 3        |
|    | 50      | North  | Own      | 6       | 25K-50K   | \$30      | 4        |
|    | 53      | North  | Rent     | 1       | 25K-50K   | \$40      | 5        |

# Lab #1 고객 데이터 분석 실습

- Q2. Show all information for only customers who (a) live in the North region, (b) own their homes, and (c) have only one household member. 북쪽에 살면서 자기집을 소유하고, 1명의 가정구성원을 가진 고객의 정보를 보여주세요.

|      |                                     |                                     |                                     |                                     |
|------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| 필드:  | CUST ID                             | REGION                              | RENT VS# OWN                        | NUM HOUSEHOLD                       |
| 테이블: | customer                            | customer                            | customer                            | customer                            |
| 정렬:  |                                     |                                     |                                     |                                     |
| 표시:  | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |
| 조건:  |                                     | "North"                             | "OWN"                               | 1                                   |



|    |         |        |            |          |           |           |          |
|----|---------|--------|------------|----------|-----------|-----------|----------|
| Q2 | CUST ID | REGION | RENT VS# ( | NUM HOU! | ANNUAL IN | TOTAL PUR | NUM PURC |
|    | 1       | North  | Own        | 1        | 10K-25K   | \$38      | 1        |
|    | 11      | North  | Own        | 1        | 50K-100K  | \$21      | 4        |
|    | 348     | North  | Own        | 1        | 50K-100K  | \$29      | 6        |
|    | 484     | North  | Own        | 1        | 25K-50K   | \$61      | 6        |

# Lab #1 고객 데이터 분석 실습

- Q3. Show all information for customers who have at least 4 household members. 최소 4명의 가족구성원을 가진 고객들의 정보를 보여주세요.
- Q4. Show all information for customers who (a) have spent less than \$20 or (b) more than \$100. 20불미만 100불이상 소비한 고객들의 정보를 보여주세요.
- Q5. Provide a two-dimensional table using crosstab that counts the number of customers by the categories of REGION and RENT VS. OWN. 크로스탭을 이용하여 2차원 테이블을 만들되 지역과 렌트대소유에 따른 고객들의 수를 보여주세요.

# Lab #1 고객 데이터 분석 실습

- Q6. Provide a two-dimensional table using crosstab that sums the TOTAL PURCHASES of customers by the categories of REGION and RENT VS. OWN. 크로스탭을 이용하여 2차원 테이블을 만들되 지역과 렌트대소유에 따른 고객들의 총구매액의 합을 보여주세요.
- Q7. Provide a query using Totals that counts the number of customers by the categories of REGION, RENT VS. OWN, and NUM HOUSEHOLD. 토탈스를 이용하여 쿼리를 만들되 지역, 렌트대소유, 가족구성원의 수에 따른 고객들의 수를 보여주세요.
- Q8-Q10. Create three more queries to answer your questions on this database. 추가적으로 3개 쿼리를 더 만드세요.

# Denormalized Data 비정규화된 데이터

- Denormalization 비정규화
  - repeat the same data wherever required. 데이터가 중복됨

| CustomerId | CustomerName | CityName |
|------------|--------------|----------|
| 1          | Sukesh       | Mumbai   |
| 2          | Rajesh       | Mumbai   |
| 3          | Mahesh       | Kolkata  |
| 4          | Ganesh       | Mumbai   |

데이터  
창고

# Normalized Data 정규화된 데이터

RDB

- Normalization 정규화
  - A database designing technique which makes sure that there will not be any redundant data.  
중복되는 데이터가 없어야 함
- Example 예

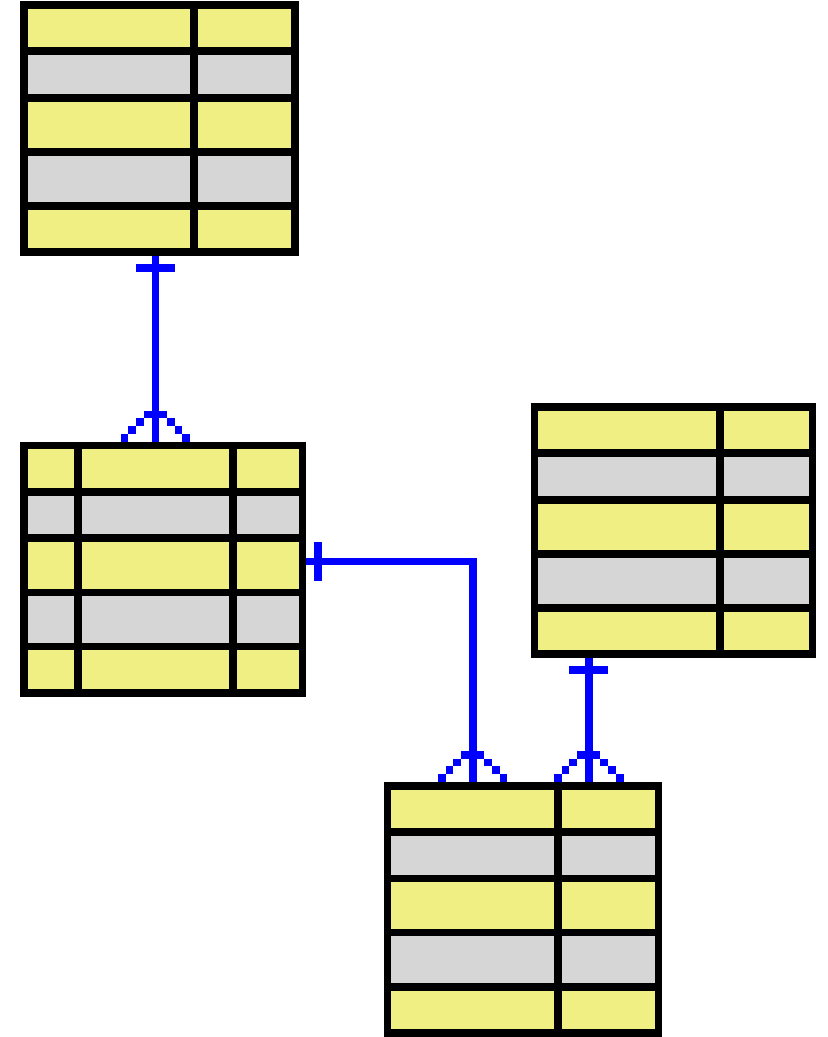
| CityId | CityName |
|--------|----------|
| 1      | Mumbai   |
| 2      | Kolkata  |
| 3      | Delhi    |

| CustomerId | CustomerName | CityId |
|------------|--------------|--------|
| 1          | Sukesh       | 1      |
| 2          | Rajesh       | 1      |
| 3          | Mahesh       | 2      |
| 4          | Ganesh       | 1      |

# Relational Database

## 관계형 데이터베이스

- Most popular database model. 가장 많이 쓰이는 데이터베이스 모델
- Series of logically related two-dimensional tables or files for storing information. 논리적으로 연관된 2차원 테이블들이나 파일들의 집합
- Tables within a relational database are related to each other. 관계형 데이터베이스 안에 있는 테이블들은 서로 연관되어 있음



# RDB Example 관계형 DB의 예

- For example, a Project List can be divided into a CUSTOMER Table, a PROJECT Table, and a PROJECT\_MANAGER Table. 프로젝트 리스트는 고객 테이블, 프로젝트 테이블, 프로젝트관리자 테이블로 나눌 수 있음
- A relational database breaks up a list into several tables—one table for each theme to minimize data redundancy while preserving complex relationships among themes. 하나의 리스트를 여러개의 테이블로 쪼갬. 데이터 중복을 최소화하기 위해 각 테이블에는 한가지 주제만 집어넣고 관계로 연결

# Table 테이블

- A record is composed of fields and contains all the data about one particular person, company, or item in a database. 레코드는 필드들로 구성되어 있고, 데이터베이스에 있는 한 개인, 회사, 아이템에 대한 모든 정보를 포함

Clients

| Client ID | Client First Name | Client Last Name | Client City | << other fields >> |
|-----------|-------------------|------------------|-------------|--------------------|
| 9001      | Stewart           | Jameson          | Seattle     | .....              |
| 9002      | Shannon           | McLain           | Poulsbo     | .....              |
| 9003      | Estela            | Pundt            | Tacoma      | .....              |
| 9004      | Timothy           | Ennis            | Seattle     | .....              |
| 9005      | Marvin            | Russo            | Bellingham  | .....              |
| 9006      | Kendra            | Bonnicksen       | Tacoma      | .....              |

Records

Fields

- A field is part of a record and contains a single piece of data for the subject of the record. 필드는 레코드의 일부로 그 레코드에 대한 정보들을 포함

# Primary Key 기본키

- Field (or group of fields) that uniquely describes each record.  
각 레코드를 구분하게 해주는 필드
- Order Number is the primary key for Order  
and Customer Number is the primary key for Customer.  
주문테이블에는 주문번호,  
고객테이블에는 고객번호 등

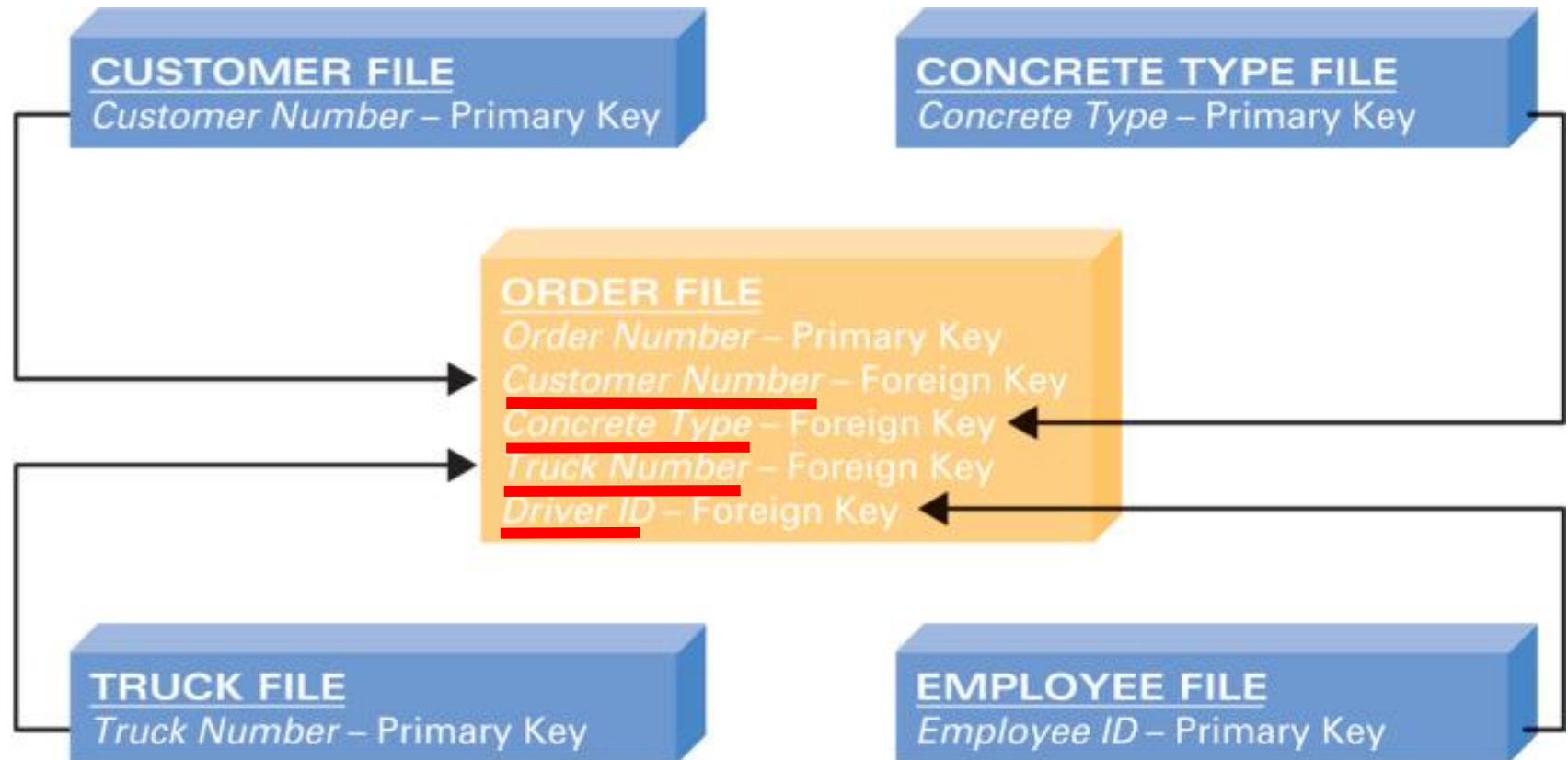
| Order Number | Order Date | Customer Number | Delivery Address  | Concrete Type | Amount | Truck Number | Driver ID |
|--------------|------------|-----------------|-------------------|---------------|--------|--------------|-----------|
| 100000       | 9/1/2004   | 1234            | 55 Smith Lane     | 1             | 8      | 111          | 123456789 |
| 100001       | 9/1/2004   | 3456            | 2122 E. Biscayne  | 1             | 3      | 222          | 785934444 |
| 100002       | 9/2/2004   | 1234            | 55 Smith Lane     | 5             | 6      | 222          | 435296657 |
| 100003       | 9/3/2004   | 4567            | 1333 Burr Ridge   | 2             | 4      | 333          | 435296657 |
| 100004       | 9/4/2004   | 4567            | 1333 Burr Ridge   | 2             | 8      | 222          | 785934444 |
| 100005       | 9/4/2004   | 5678            | 1222 Westminster  | 1             | 4      | 222          | 785934444 |
| 100006       | 9/5/2004   | 1234            | 222 East Hampton  | 1             | 4      | 111          | 123456789 |
| 100007       | 9/6/2004   | 2345            | 9 W. Palm Beach   | 2             | 5      | 333          | 785934444 |
| 100008       | 9/6/2004   | 6789            | 4532 Lane Circle  | 1             | 8      | 222          | 785934444 |
| 100009       | 9/7/2004   | 1234            | 987 Furlong       | 3             | 8      | 111          | 123456789 |
| 100010       | 9/9/2004   | 6789            | 4532 Lance Circle | 2             | 7      | 222          | 435296657 |
| 100011       | 9/9/2004   | 4567            | 3500 Tomahawk     | 5             | 6      | 222          | 785934444 |

| Customer Number | Customer Name          | Customer Phone | Customer Primary Contact |
|-----------------|------------------------|----------------|--------------------------|
| 1234            | Smelding Homes         | 3333333333     | Bill Johnson             |
| 2345            | Home Builders Superior | 3334444444     | Marcus Connolly          |
| 3456            | Mark Akey              | 3335555555     | Mark Akey                |
| 4567            | Triple A Homes         | 3336666666     | Janielle Smith           |
| 5678            | Sheryl Williamson      | 3337777777     | Sheryl Williamson        |
| 6789            | Home Makers            | 3338888888     | John Yu                  |

# Foreign Key 외래키

- Primary key of one file that appears in another file. 기본키가 다른 테이블에 나타나면 외래키
- Customer Number appears in Order as a foreign key. 주문 테이블에 있는 고객번호는 외래키

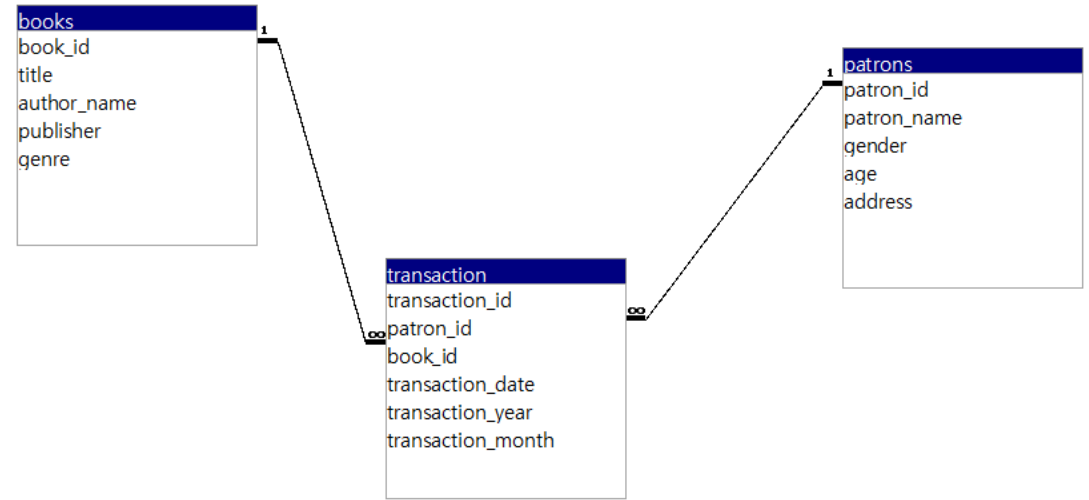


# Referential Integrity 참조무결성

- Rules that help ensure the quality of information. 정보의 질을 유지하기 위한 방법
- Data dictionary, for example, defines type of information - numeric, date, and so on. 데이터 정의를 이용하여 데이터 타입을 정함. 데이터 타입이 안맞으면 에러!!!
- Foreign keys - must be found as primary keys in another file. 외래키로 사용하기 위해서는 다른 테이블의 기본키여야 함
- E.g., a Customer Number in the Order Table must also be present in the Customer Table. 고객테이블에 고객번호가 있어야 주문테이블에서 고객번호를 사용할수 있음

# Lab #1 도서관 데이터 분석 실습

- Create a simple library database with three tables: books, patrons, and transaction. 세개의 테이블을 가진 도서관 데이터베이스를 만드시오.
  - 테이블을 임포트한 후에 merge를 이용해 합치기



# Sample SQL Code for books and patrons

```
create table books(  
    book_id char(13),  
    title varchar(100),  
    author_name varchar(50),  
    publisher varchar(50),  
    genre varchar(50),  
    constraint pk_book_id primary  
key (book_id));
```

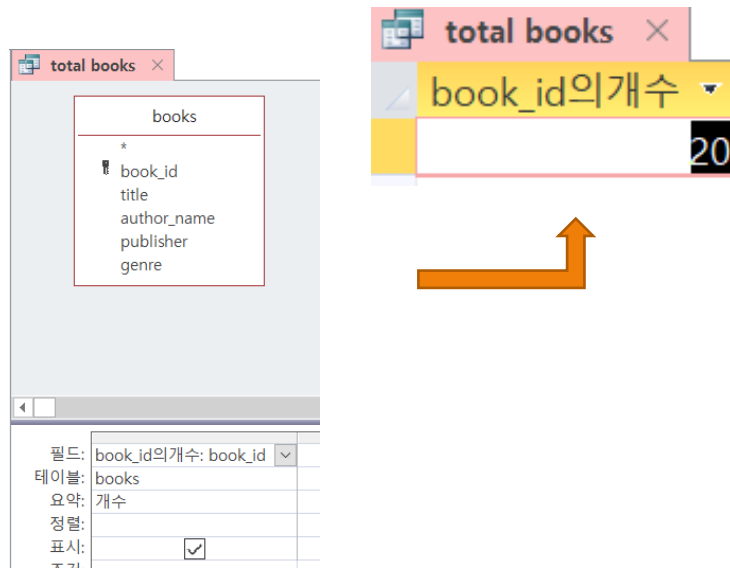
```
create table patrons(  
    patron_id char(5),  
    patron_name varchar(80),  
    gender char(1),  
    age number,  
    address varchar(200),  
    constraint pk_patron_id primary  
key (patron_id));
```

# Sample SQL Code for transaction

```
create table transaction(  
    transaction_id char(5),  
    patron_id char(5),  
    book_id char(13),  
    transaction_date char(50),  
    transaction_year char(4),  
    transaction_month varchar(2),  
    constraint pk_transaction_id primary key(transaction_id),  
    constraint fk_patron_id foreign key(patron_id) references  
patron(patron_id),  
    constraint fk_book_id foreign key(book_id) references  
book(book_id) );
```

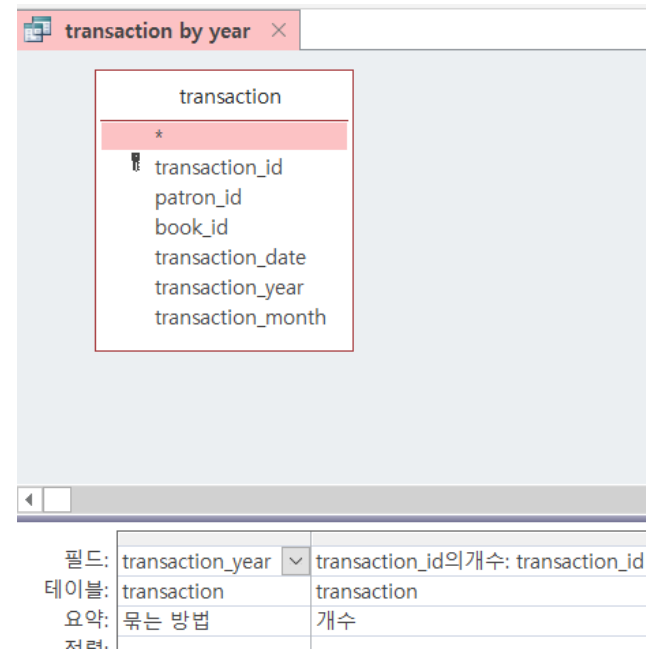
# Lab #1 도서관 데이터 분석 실습

- Create relationships among tables and answer the questions. 테이블 사이에 관계를 만들고 질문에 답하십시오.
- Total number of books? 전체 책의 숫자는?
- Total number of patrons? 총 고객수?
- Total number of transactions? 총 거래수?

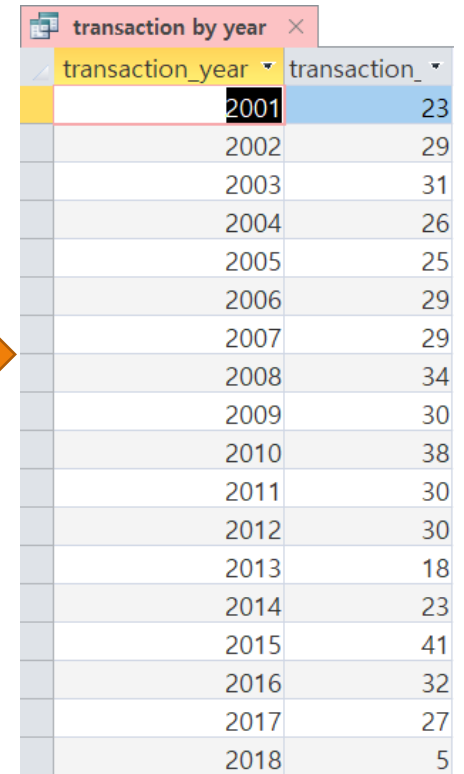
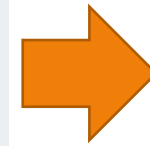


# Lab #1 도서관 데이터 분석 실습

- Number of transactions by year? 년도별 거래수?
- Number of transactions by genre? 장르별 거래수?
- Number of transactions by month? 달별 거래수?



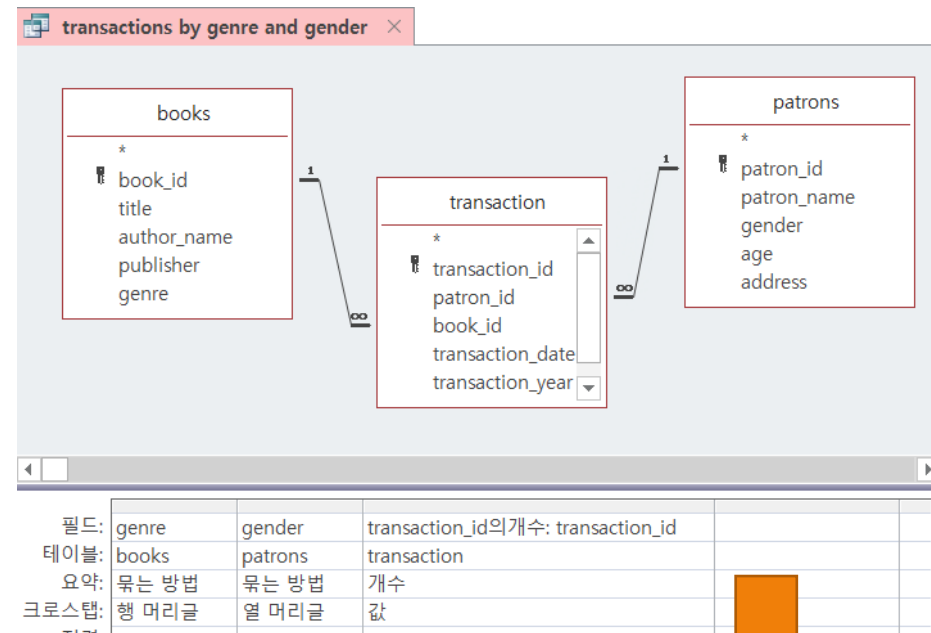
| 필드:  | transaction_year | transaction_id의개수: transaction_id |
|------|------------------|-----------------------------------|
| 테이블: | transaction      | transaction                       |
| 요약:  | 읽는 방법            | 개수                                |
| 저장:  |                  |                                   |



| transaction_year | transaction_id |
|------------------|----------------|
| 2001             | 23             |
| 2002             | 29             |
| 2003             | 31             |
| 2004             | 26             |
| 2005             | 25             |
| 2006             | 29             |
| 2007             | 29             |
| 2008             | 34             |
| 2009             | 30             |
| 2010             | 38             |
| 2011             | 30             |
| 2012             | 30             |
| 2013             | 18             |
| 2014             | 23             |
| 2015             | 41             |
| 2016             | 32             |
| 2017             | 27             |
| 2018             | 5              |

# Lab #1 도서관 데이터 분석 실습

- Number of transactions by genre and gender?  
장르별 성별 거래수?
- Number of transactions by month and gender?  
달별 성별 거래수?

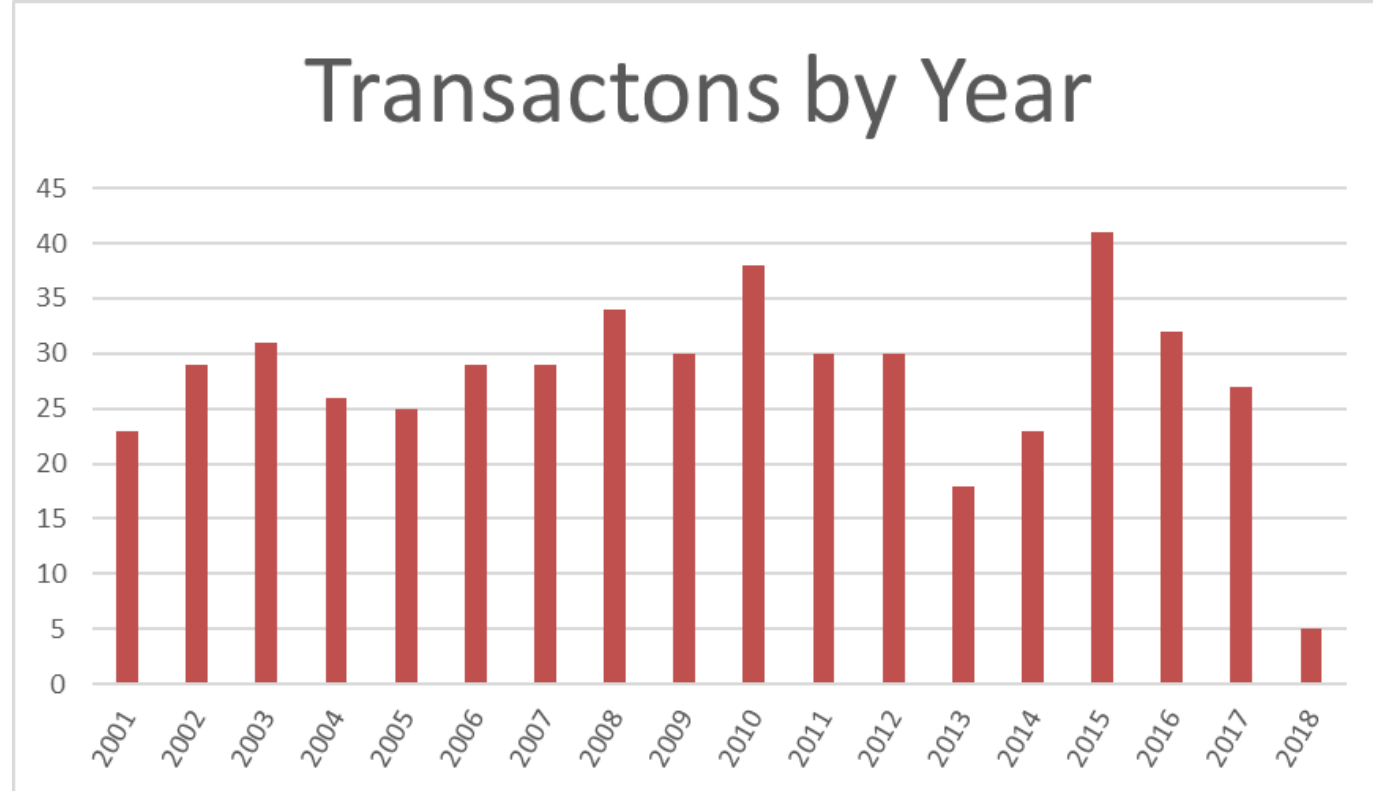


| genre | F  | M   |
|-------|----|-----|
| 소설    | 47 | 53  |
| 아동도서  | 27 | 23  |
| 에세이   | 91 | 109 |
| 학습서   | 73 | 77  |

# Number of Transactions by Year

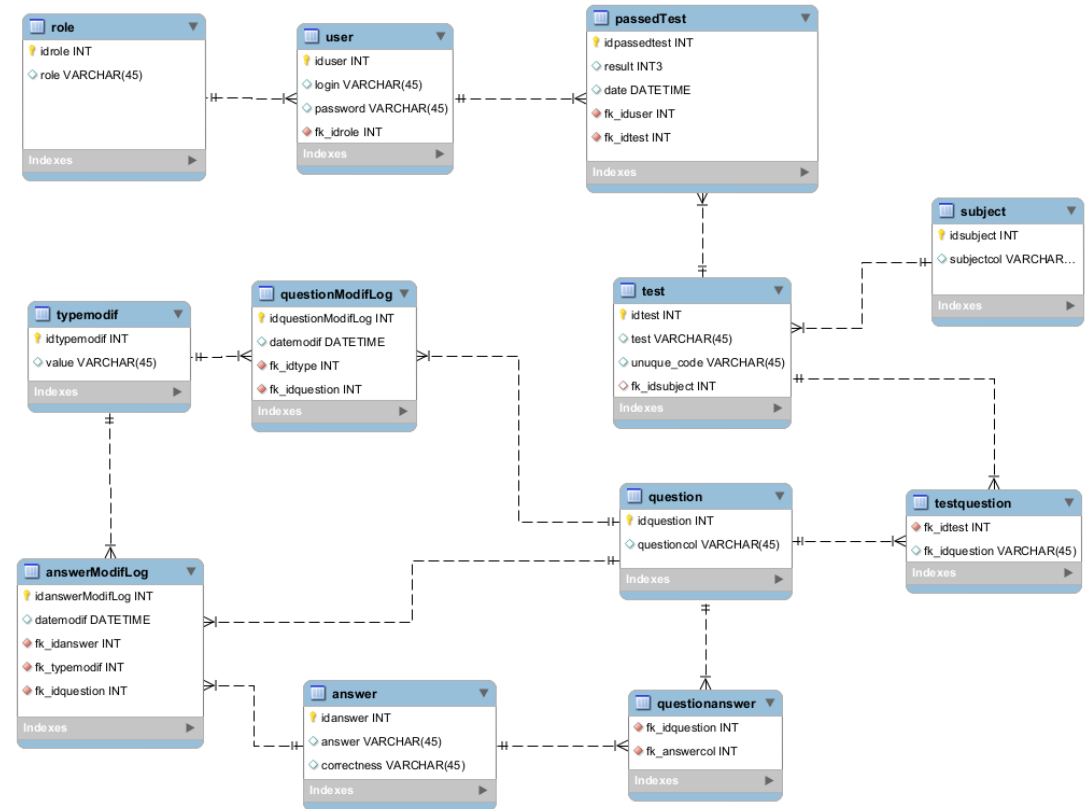
## 년도별 거래수

| transaction by year |              |
|---------------------|--------------|
| transaction_year    | transaction_ |
| 2001                | 23           |
| 2002                | 29           |
| 2003                | 31           |
| 2004                | 26           |
| 2005                | 25           |
| 2006                | 29           |
| 2007                | 29           |
| 2008                | 34           |
| 2009                | 30           |
| 2010                | 38           |
| 2011                | 30           |
| 2012                | 30           |
| 2013                | 18           |
| 2014                | 23           |
| 2015                | 41           |
| 2016                | 32           |
| 2017                | 27           |
| 2018                | 5            |



# Relational Database Management Systems (RDBMS)

- An application or collection of applications designed to create, store, manage, and facilitate access to relational databases. 관계형 데이터베이스를 생성, 저장, 관리하는 어플리케이션
- The RDBMS also manages and controls database activities. 사용자의 데이터베이스 사용 활동 관리



# Commercial DBMS Products

## 상업용 DBMS

### Desktop DBMS Products

- Microsoft Access

### Organizational DBMS Products

- Microsoft's SQL Server
- Oracle's Oracle
- Sun Microsystems's MySQL
- IBM's DB2

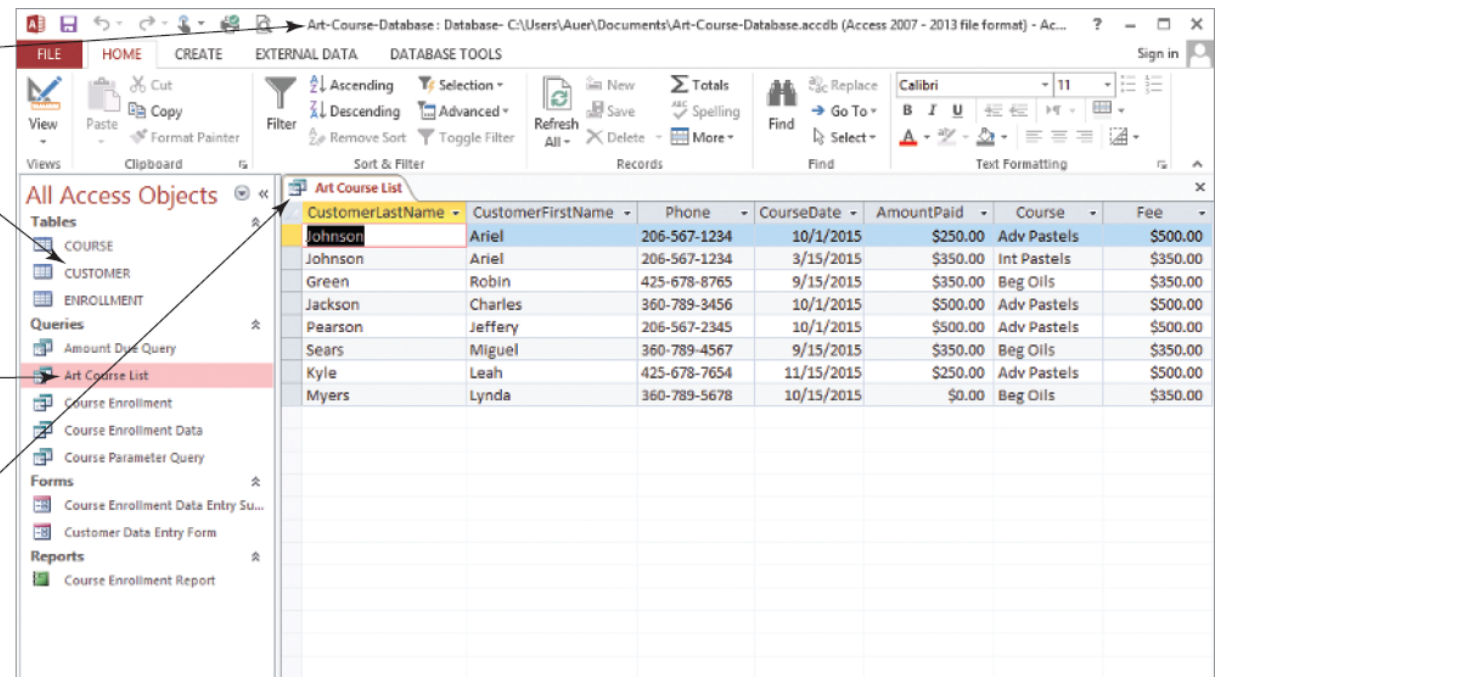
# Microsoft Access 마이크로소프트 액세스

The database name  
**Art-Course-Database**

The table object  
**CUSTOMER** is  
displayed under the All  
Access Objects

The query object  
Art Course List stores  
the query itself

The query results in  
table format



| CustomerLastName | CustomerFirstName | Phone        | CourseDate | AmountPaid | Course      | Fee      |
|------------------|-------------------|--------------|------------|------------|-------------|----------|
| Johnson          | Ariel             | 206-567-1234 | 10/1/2015  | \$250.00   | Adv Pastels | \$500.00 |
| Johnson          | Ariel             | 206-567-1234 | 3/15/2015  | \$350.00   | Int Pastels | \$350.00 |
| Green            | Robin             | 425-678-8765 | 9/15/2015  | \$350.00   | Beg Oils    | \$350.00 |
| Jackson          | Charles           | 360-789-3456 | 10/1/2015  | \$500.00   | Adv Pastels | \$500.00 |
| Pearson          | Jeffery           | 206-567-2345 | 10/1/2015  | \$500.00   | Adv Pastels | \$500.00 |
| Sears            | Miguel            | 360-789-4567 | 9/15/2015  | \$350.00   | Beg Oils    | \$350.00 |
| Kyle             | Leah              | 425-678-7654 | 11/15/2015 | \$250.00   | Adv Pastels | \$500.00 |
| Myers            | Lynda             | 360-789-5678 | 10/15/2015 | \$0.00     | Beg Oils    | \$350.00 |

# Microsoft SQL Server

## 마이크로소프트 SQL서버

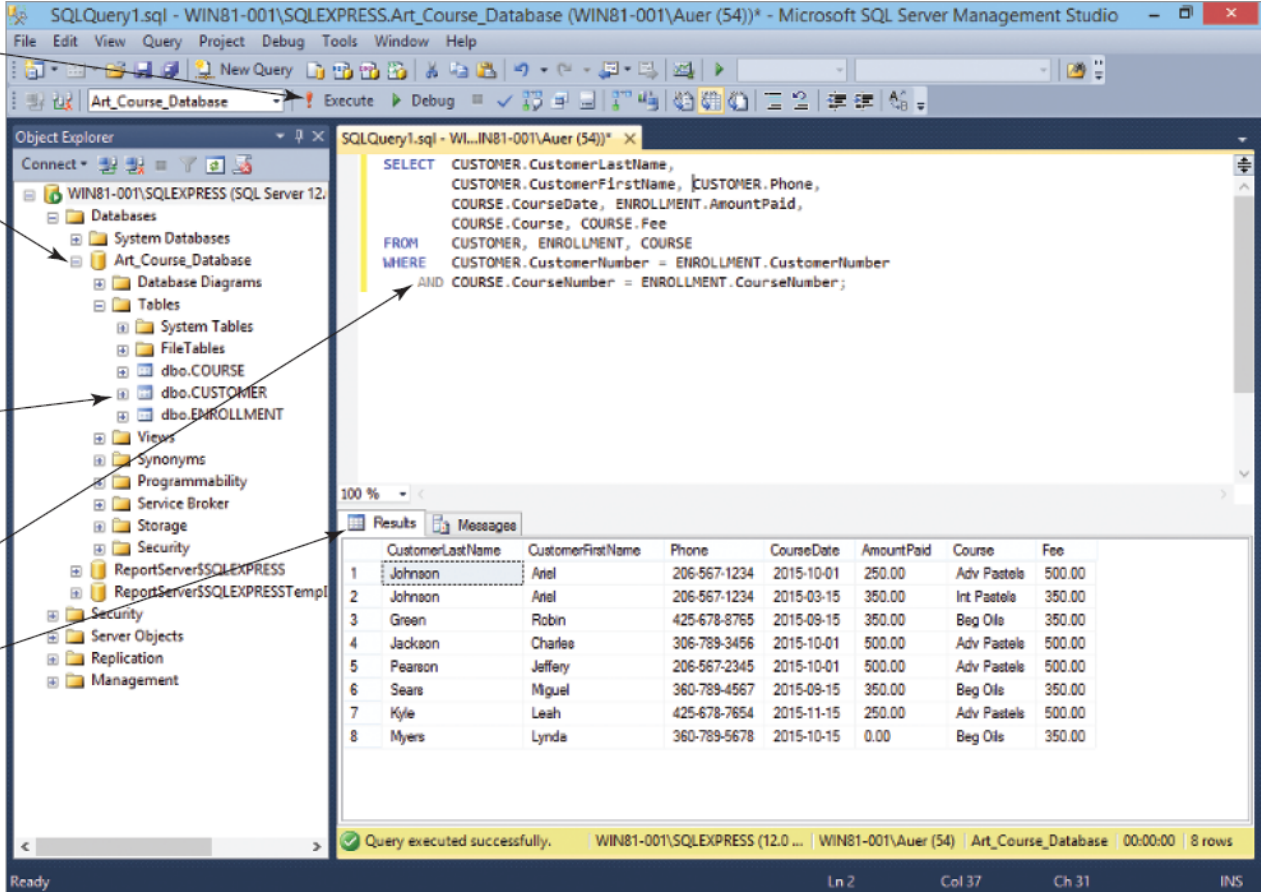
Click this button to run the SQL query

The database object **Art-Course-Database** is displayed in the Object Explorer

The table object **CUSTOMER** is displayed under the Art-Course-Database object

The SQL query

The query results in table format



The screenshot shows the Microsoft SQL Server Management Studio interface. The title bar indicates the connection to 'WIN81-001\SQLEXPRESS.Art\_Course\_Database (WIN81-001\Auer (54))'. The Object Explorer on the left shows the database structure, with 'Art\_Course\_Database' expanded to show tables 'dbo.COURSE', 'dbo.CUSTOMER', and 'dbo.ENROLLMENT'. The SQL query editor in the center contains a SELECT statement joining CUSTOMER, ENROLLMENT, and COURSE tables. The Results pane at the bottom displays the query output in a table format with 8 rows. A status bar at the bottom indicates the query was executed successfully, returning 8 rows.

```
SELECT CUSTOMER.CustomerLastName,
CUSTOMER.CustomerFirstName, CUSTOMER.Phone,
COURSE.CourseDate, ENROLLMENT.AmountPaid,
COURSE.Course, COURSE.Fee
FROM CUSTOMER, ENROLLMENT, COURSE
WHERE CUSTOMER.CustomerNumber = ENROLLMENT.CustomerNumber;
AND COURSE.CourseNumber = ENROLLMENT.CourseNumber;
```

|   | CustomerLastName | CustomerFirstName | Phone        | CourseDate | AmountPaid | Course      | Fee    |
|---|------------------|-------------------|--------------|------------|------------|-------------|--------|
| 1 | Johnson          | Ariel             | 206-567-1234 | 2015-10-01 | 250.00     | Adv Pastels | 500.00 |
| 2 | Johnson          | Ariel             | 206-567-1234 | 2015-03-15 | 350.00     | Int Pastels | 350.00 |
| 3 | Green            | Robin             | 425-678-8765 | 2015-09-15 | 350.00     | Beg Oils    | 350.00 |
| 4 | Jackson          | Charles           | 360-789-3456 | 2015-10-01 | 500.00     | Adv Pastels | 500.00 |
| 5 | Pearson          | Jeffery           | 206-567-2345 | 2015-10-01 | 500.00     | Adv Pastels | 500.00 |
| 6 | Sears            | Miguel            | 360-789-4567 | 2015-09-15 | 350.00     | Beg Oils    | 350.00 |
| 7 | Kyle             | Leah              | 425-678-7654 | 2015-11-15 | 250.00     | Adv Pastels | 500.00 |
| 8 | Myers            | Lynda             | 360-789-5678 | 2015-10-15 | 0.00       | Beg Oils    | 350.00 |

Query executed successfully. WIN81-001\SQLEXPRESS (12.0 ... | WIN81-001\Auer (54) | Art\_Course\_Database | 00:00:00 | 8 rows

# Oracle Database Express Edition 11g Release 2 오라클 데이터베이스 익스프레스 에디션

The database object **Art-Course-Database** is displayed in the Connections browser

Click this button to run the SQL query

The table object **CUSTOMER** is displayed in the Tables objects

The SQL query

The query results in table format

The screenshot shows the Oracle SQL Developer interface with the 'Art\_Course\_Database' connection selected. The 'Connections' browser on the left shows the database structure, including the 'CUSTOMER' table. The 'Worksheet' tab displays the following SQL query:

```
SELECT CUSTOMER.CustomerLastName,
CUSTOMER.CustomerFirstName, CUSTOMER.Phone,
COURSE.CourseDate, ENROLLMENT.AmountPaid,
COURSE.Course, COURSE.Fee
FROM CUSTOMER, ENROLLMENT, COURSE
WHERE CUSTOMER.CustomerNumber = ENROLLMENT.CustomerNumber
AND COURSE.CourseNumber = ENROLLMENT.CourseNumber;
```

The 'Query Result' tab shows the results in table format:

|   | CUSTOMERLASTNAME | CUSTOMERFIRSTNAME | PHONE        | COURSEDATE | AMOUNTPAID | COURSE      | FEE |
|---|------------------|-------------------|--------------|------------|------------|-------------|-----|
| 1 | Johnson          | Ariel             | 206-567-1234 | 15-MAR-15  | 350        | Int Pastels | 350 |
| 2 | Johnson          | Ariel             | 206-567-1234 | 01-OCT-15  | 250        | Adv Pastels | 500 |
| 3 | Green            | Robin             | 425-678-8765 | 15-SEP-15  | 350        | Beg Oils    | 350 |
| 4 | Jackson          | Charles           | 306-789-3456 | 01-OCT-15  | 500        | Adv Pastels | 500 |
| 5 | Pearson          | Jeffery           | 206-567-2345 | 01-OCT-15  | 500        | Adv Pastels | 500 |
| 6 | Sears            | Miguel            | 360-789-4567 | 15-SEP-15  | 350        | Beg Oils    | 350 |
| 7 | Kyle             | Leah              | 425-678-7654 | 15-NOV-15  | 250        | Adv Pastels | 500 |
| 8 | Myers            | Lynda             | 360-789-5678 | 15-OCT-15  | 0          | Beg Oils    | 350 |

Oracle SQL Developer : Art\_Course\_Database

File Edit View Navigate Run Source Team Tools Window Help

Connections

Art\_Course\_Database

Tables (Filtered)

COURSE

CUSTOMER

ENROLLMENT

Views

Editing Views

Indexes

Packages

Procedures

Functions

Queues

Queues Tables

Triggers

Crossedition Triggers

Types

Sequences

Materialized Views

Materialized View Logs

Synonyms

Public Synonyms

Database Links

Public Database Links

Directories

Reports

All Reports

Data Dictionary Reports

Data Modeler Reports

OLAP Reports

TimesTen Reports

User Defined Reports

Worksheet

Query Builder

Query Result

All Rows Fetched: 8 in 0.094 seconds

Line 1 Column 1 | Insert | Modified | Windows: O

# MySQL 5.6

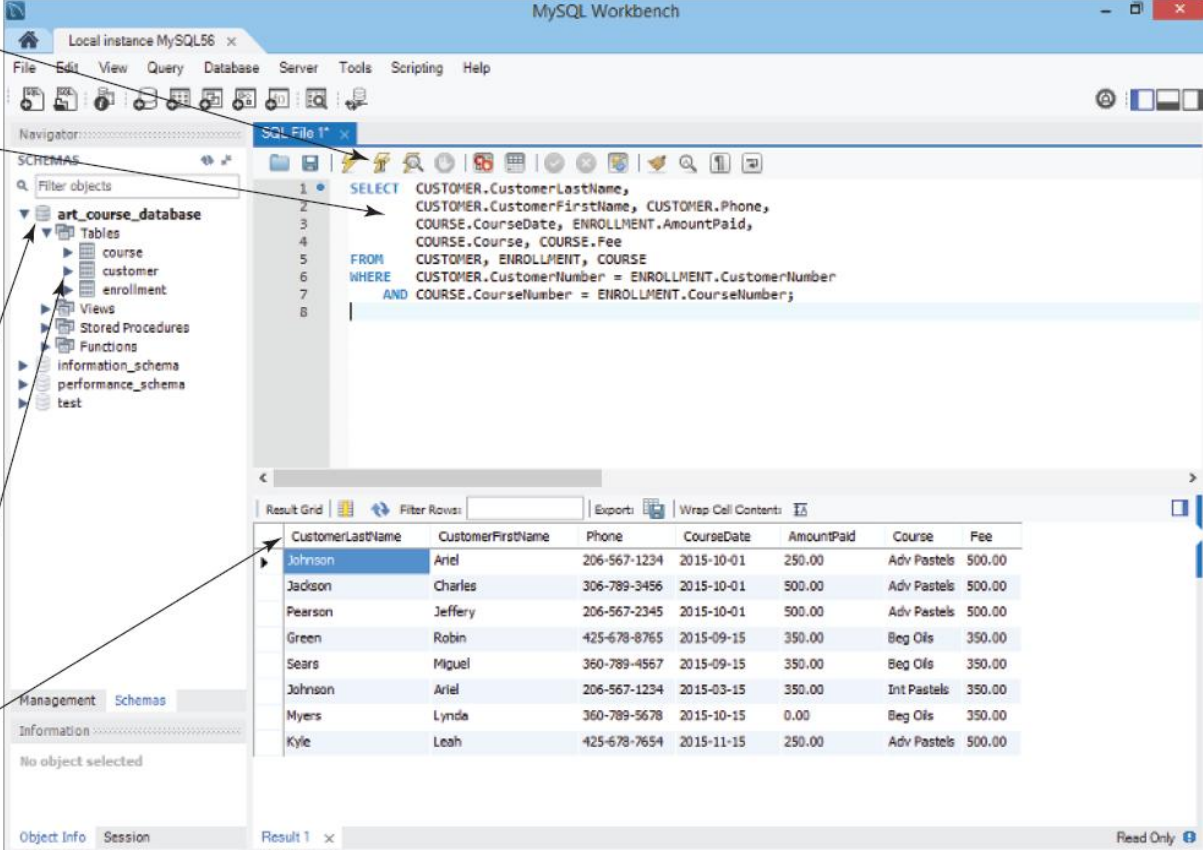
Click this button to run the SQL query

The SQL query

The database object **art\_course\_database** is displayed in the Object Browser

The table object **CUSTOMER** is displayed under the art-course-database object

The query results in table format



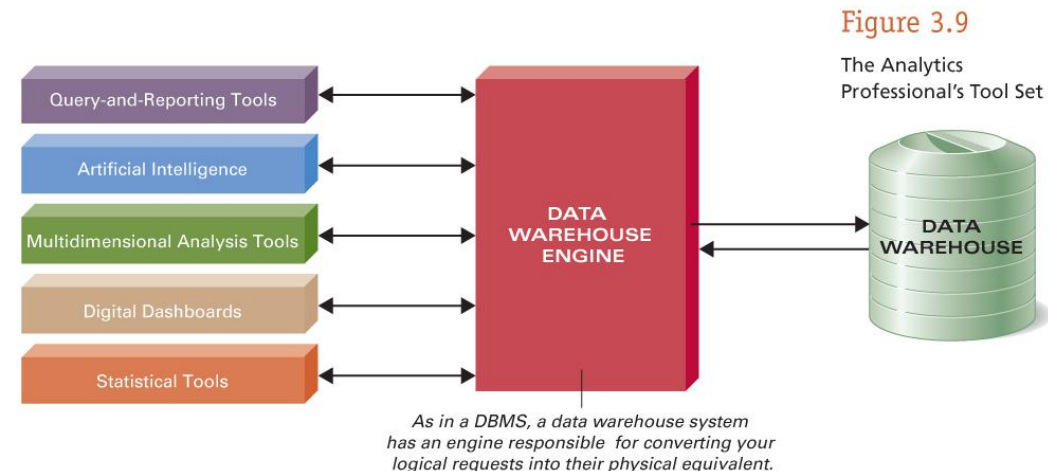
The screenshot shows the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The left sidebar contains the Navigator and Object Browser. The Object Browser shows the hierarchy: Schemas > art\_course\_database > Tables > customer. The main window displays a SQL query in a text editor. The bottom pane shows the results of the query in a table format.

```
SELECT CUSTOMER.CustomerLastName,
CUSTOMER.CustomerFirstName, CUSTOMER.Phone,
COURSE.CourseDate, ENROLLMENT.AmountPaid,
COURSE.Course, COURSE.Fee
FROM CUSTOMER, ENROLLMENT, COURSE
WHERE CUSTOMER.CustomerNumber = ENROLLMENT.CustomerNumber
AND COURSE.CourseNumber = ENROLLMENT.CourseNumber;
```

| CustomerLastName | CustomerFirstName | Phone        | CourseDate | AmountPaid | Course      | Fee    |
|------------------|-------------------|--------------|------------|------------|-------------|--------|
| Johnson          | Ariel             | 206-567-1234 | 2015-10-01 | 250.00     | Adv Pastels | 500.00 |
| Jackson          | Charles           | 306-789-3456 | 2015-10-01 | 500.00     | Adv Pastels | 500.00 |
| Pearson          | Jeffery           | 206-567-2345 | 2015-10-01 | 500.00     | Adv Pastels | 500.00 |
| Green            | Robin             | 425-678-8765 | 2015-09-15 | 350.00     | Beg Oils    | 350.00 |
| Sears            | Miguel            | 360-789-4567 | 2015-09-15 | 350.00     | Beg Oils    | 350.00 |
| Johnson          | Ariel             | 206-567-1234 | 2015-03-15 | 350.00     | Int Pastels | 350.00 |
| Myers            | Lynda             | 360-789-5678 | 2015-10-15 | 0.00       | Beg Oils    | 350.00 |
| Kyle             | Leah              | 425-678-7654 | 2015-11-15 | 250.00     | Adv Pastels | 500.00 |

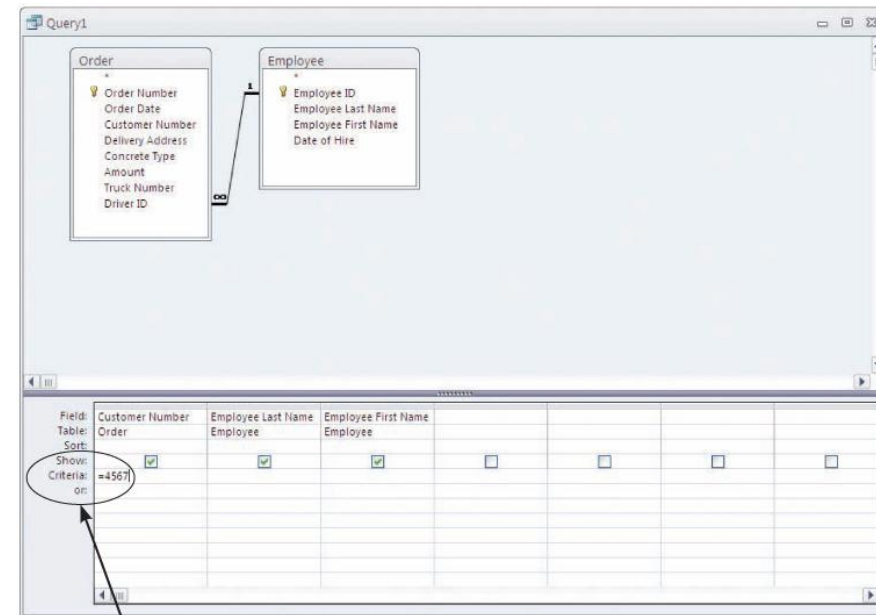
# Data-Mining Tools 데이터 마이닝 툴들

- Software tools you use in a data warehouse environment. 데이터 창고에서 사용하는 소프트웨어 툴들
  - Query-and-reporting tools
  - Artificial intelligence
  - Multidimensional analysis tools
  - Digital dashboards
  - Statistical tools



# Query-by-Example (QBE) Tool

- You can graphically design the answer to a question. 질문에 답하기 위해 그래픽하게 설계



Our selection criteria

| Customer Number | Employee Last Name | Employee First Name |
|-----------------|--------------------|---------------------|
| 4567            | Evaraz             | Antonio             |
| 4567            | Robertson          | John                |
| 4567            | Robertson          | John                |
| 4567            | Robertson          | John                |

Figure 3.7

Using a Query-by-Example to Find Information

# Report Generator 리포트 생성기

- You can quickly define formats of reports and what information you want to see in a report. 리포트의 형식, 무슨 정보를 넣을 지를 빠르게 정의할 수 있음

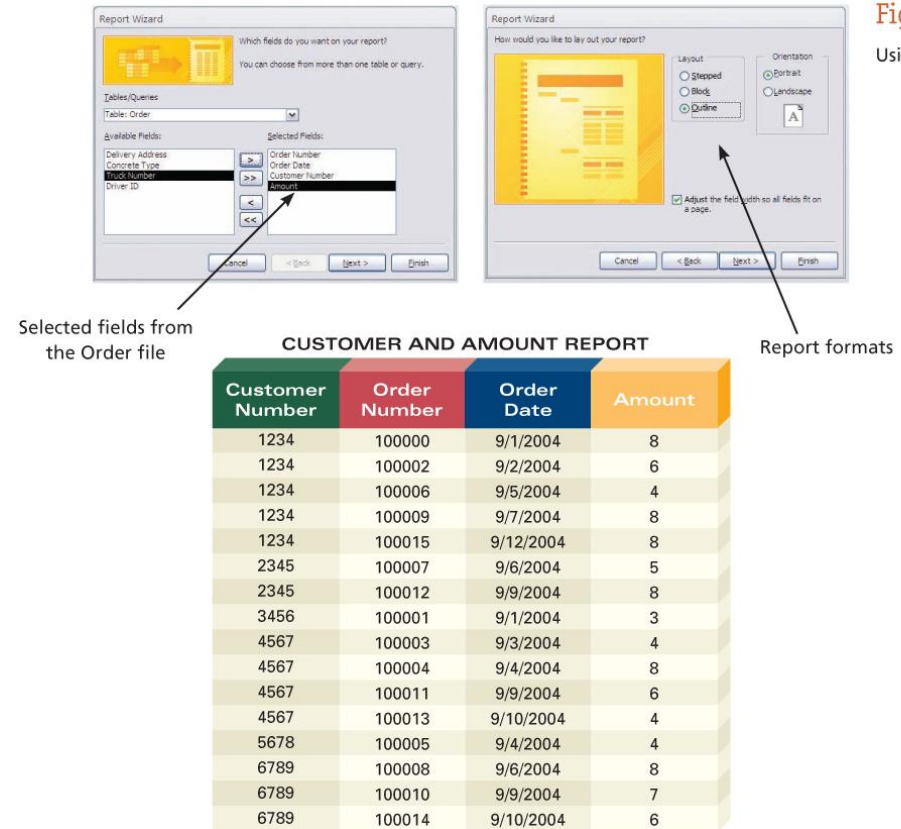


Figure 3.6  
Using a Report Generator

# Structured Query Language (SQL)

## 구조적 질의어

- An international standard for creating, processing, and querying databases and their tables.
- 데이터베이스와 테이블들을  
생성, 처리, 질의 하는 국제표준

```
SELECT CUSTOMER.CustomerLastName,  
        CUSTOMER.CustomerFirstName,  
        CUSTOMER.Phone,  
        COURSE.CourseDate,  
        ENROLLMENT.AmountPaid,  
        COURSE.Course, COURSE.Fee  
FROM    CUSTOMER, ENROLLMENT, COURSE  
WHERE   CUSTOMER.CustomerNumber  
        =  
        ENROLLMENT.CustomerNumber  
        AND   COURSE.CourseNumber  
        =  
        ENROLLMENT.CourseNumber;
```

# Artificial Intelligence 인공지능

- Tools to help you “discover” information and trends. 정보와 추세의 발견을 돕는 툴들

## Artificial Intelligence

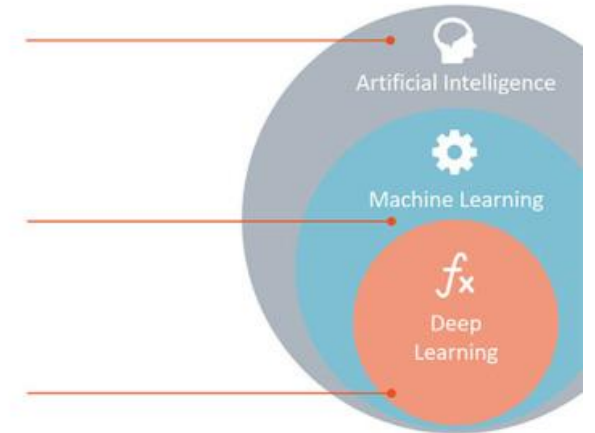
Any technique which enables computers to mimic human behavior.

## Machine Learning

Subset of AI techniques which use statistical methods to enable machines to improve with experiences.

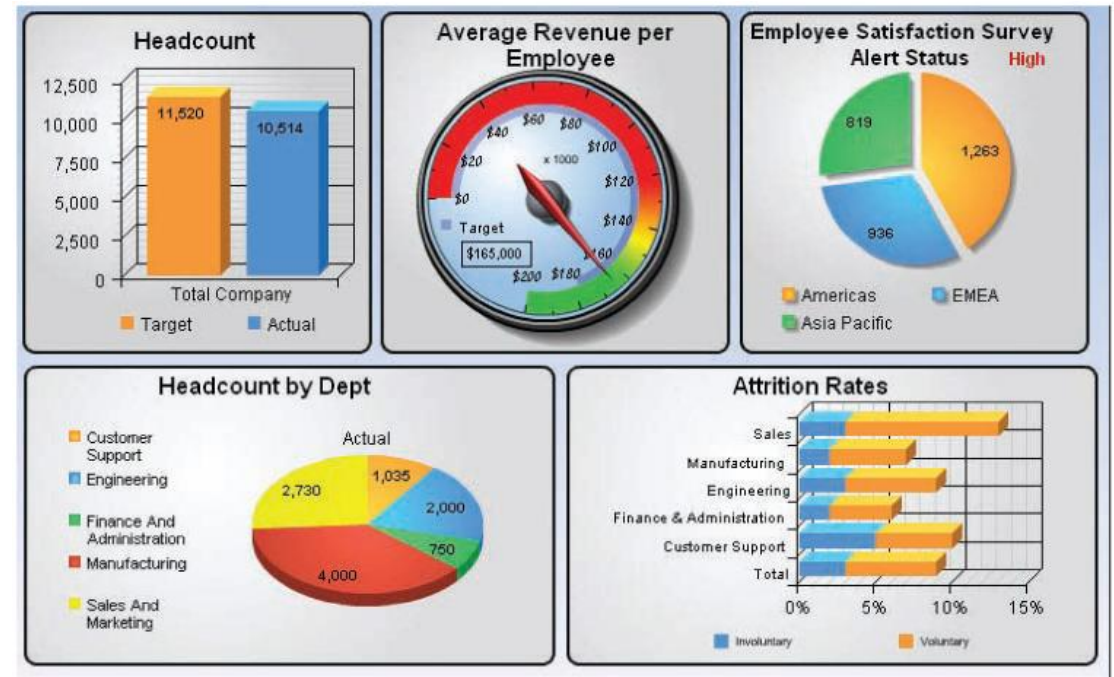
## Deep Learning

Subset of ML which make the computation of multi-layer neural networks feasible.



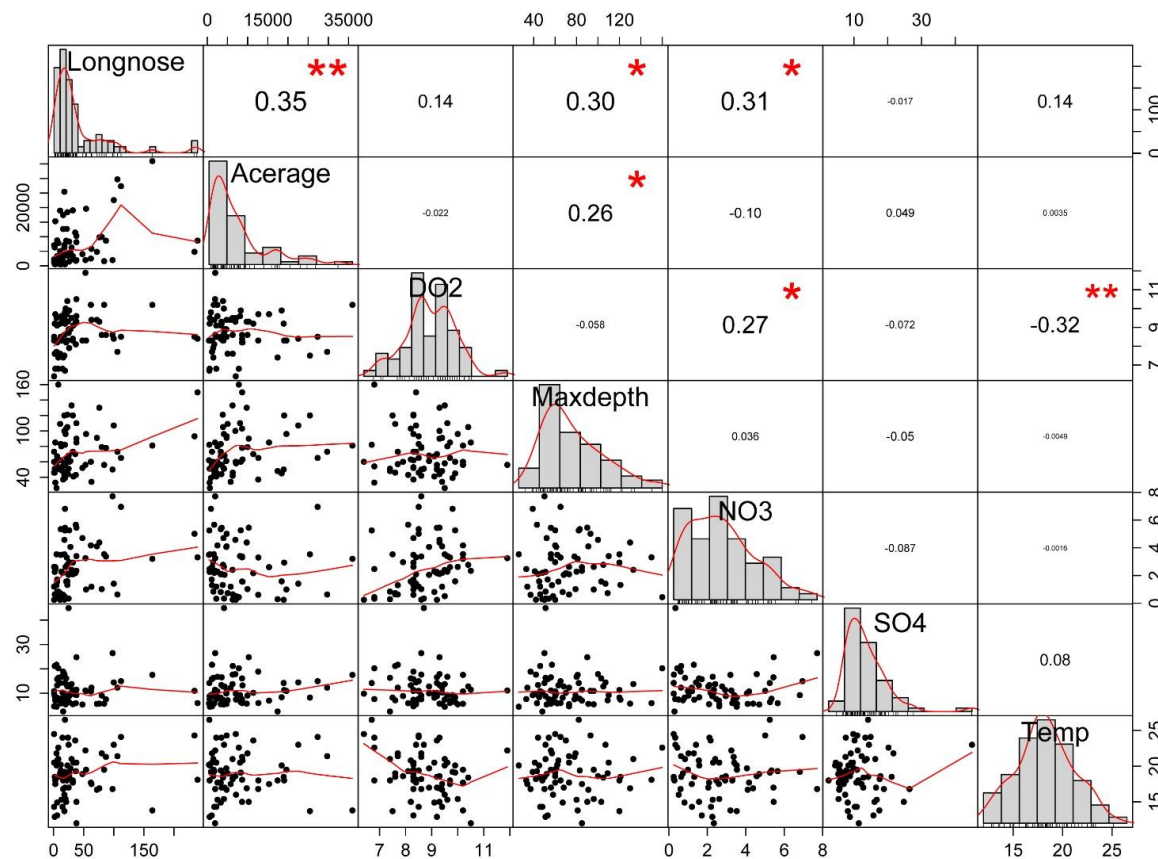
# Digital Dashboard 대시보드

- Displays key information on a computer screen tailored to the needs and wants of an individual. 개인의 필요에 맞춰 주요 정보를 보여줌
- Key performance indicator (KPI) - most essential information used in any analytics initiative. 주요 성과 지표는 분석에 사용되는 가장 중요한 정보



# Statistical Tools 통계툴

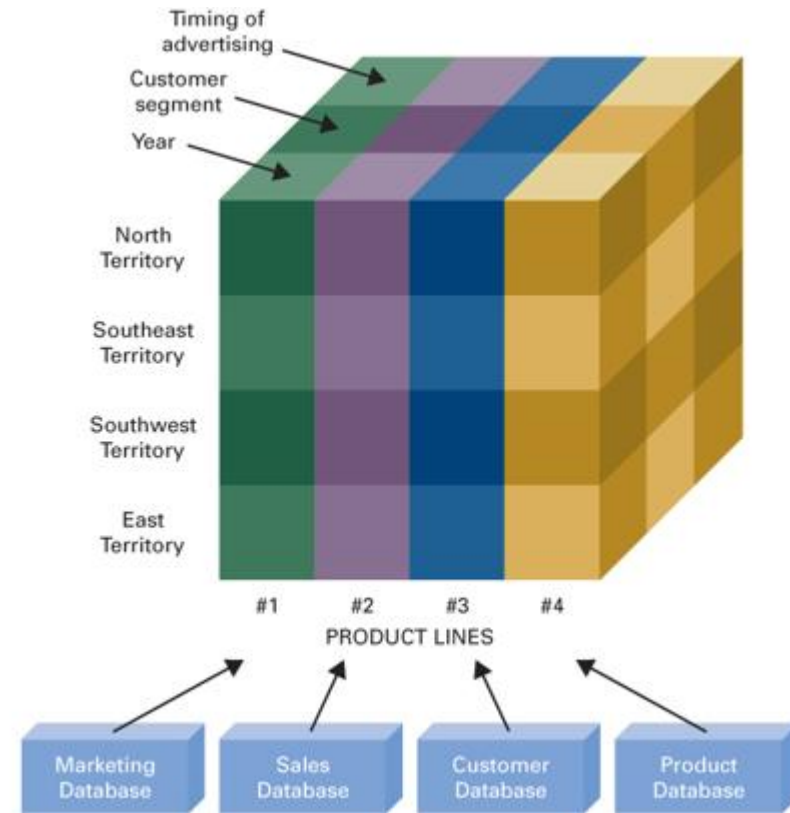
- Applying mathematical models to data warehouse information.  
e.g., linear regression.  
데이터창고에 선형회귀와 같은 수학적모델을 적용



# Multidimensional Analysis (MDA)

## Tools

- Slice-and-dice techniques for viewing multidimensional information. 다차원 정보를 보기 위해 잘라서 큐브를 만드는 기술



# DB-Engines Ranking

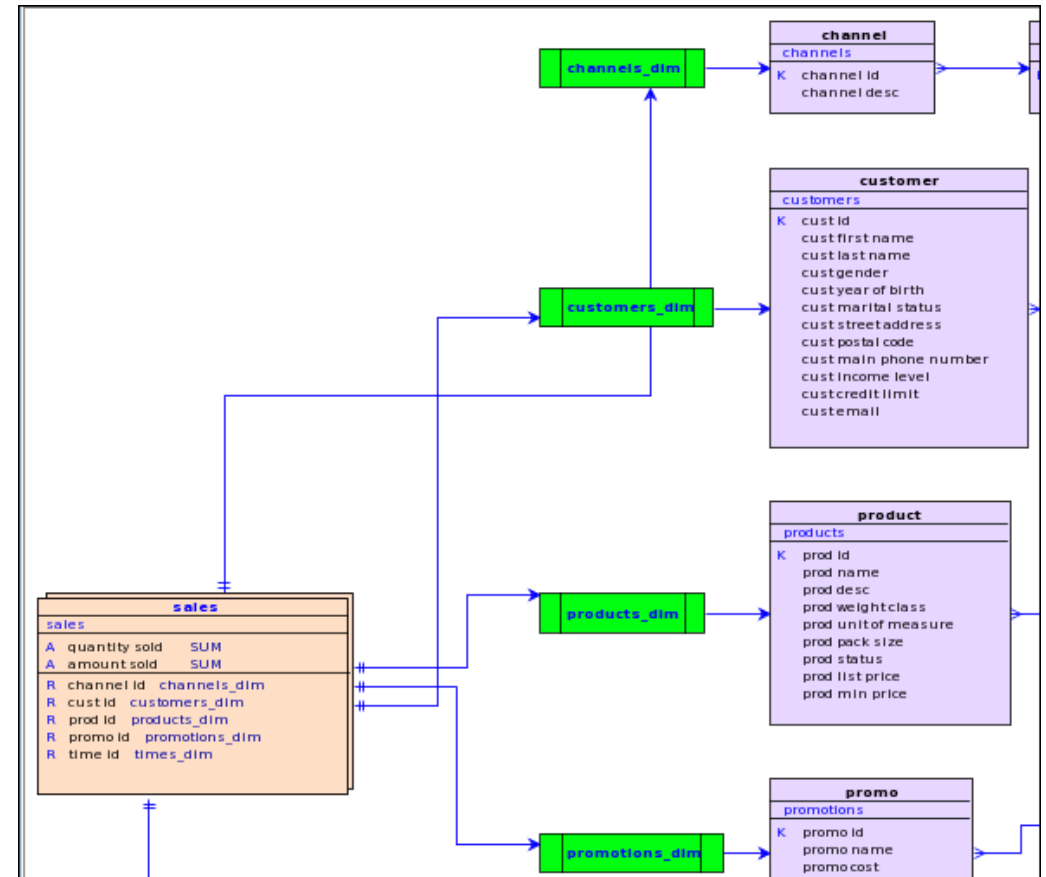
## 데이터베이스 엔진 순위

| Rank     |          |          | DBMS          | Database Model            |
|----------|----------|----------|---------------|---------------------------|
| Sep 2026 | Aug 2026 | Sep 2025 |               |                           |
| 1.       | 1.       | 1.       | Oracle        | Relational, Multi-model ⓘ |
| 2.       | 2.       | 2.       | MySQL         | Relational, Multi-model ⓘ |
|          |          |          | Microsoft SQL |                           |
| 3.       | 3.       | 3.       | Server        | Relational, Multi-model ⓘ |
| 4.       | 4.       | 4.       | PostgreSQL    | Relational, Multi-model ⓘ |
| 5.       | 5.       | 5.       | MongoDB       | Document, Multi-model ⓘ   |
| 6.       | 6.       | 6.       | Snowflake     | Relational                |
| 7.       | 7.       | ↑ 9.     | Databricks    | Multi-model ⓘ             |
| 8.       | 8.       | ↓ 7.     | Redis         | Key-value, Multi-model ⓘ  |
| 9.       | 9.       | ↓ 8.     | IBM Db2       | Relational, Multi-model ⓘ |
| 10.      | ↑ 11.    | ↑ 11.    | SQLite        | Relational                |

# Multidimensional Data Model

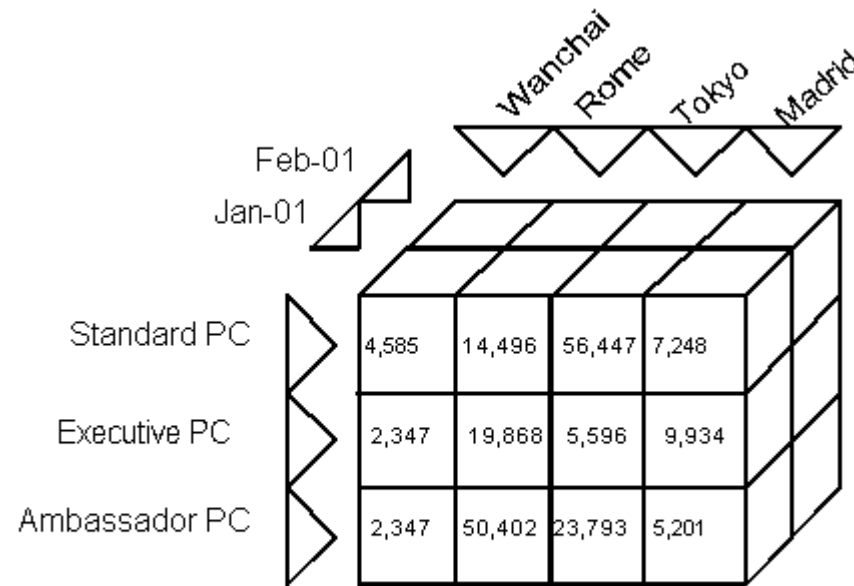
## 다차원 데이터 모델

- An integral part of OLAP (Online Analytical Processing). OLAP을 통합함
- Designed to solve complex queries in real time. 실시간으로 복잡한 쿼리를 해결하도록 디자인
- Composed of logical cubes, measures, dimensions, hierarchies, levels, attributes. 논리적 큐브, 측정치, 차원, 계층, 레벨, 속성으로 구성



# Cube 큐브

- Organizes measures that have the exact same dimensions. 측정치를 같은 차원으로 구성
- Can easily be analyzed and displayed together. 쉽게 분석하고 함께 보여줄 수 있음



# Measures 측정치

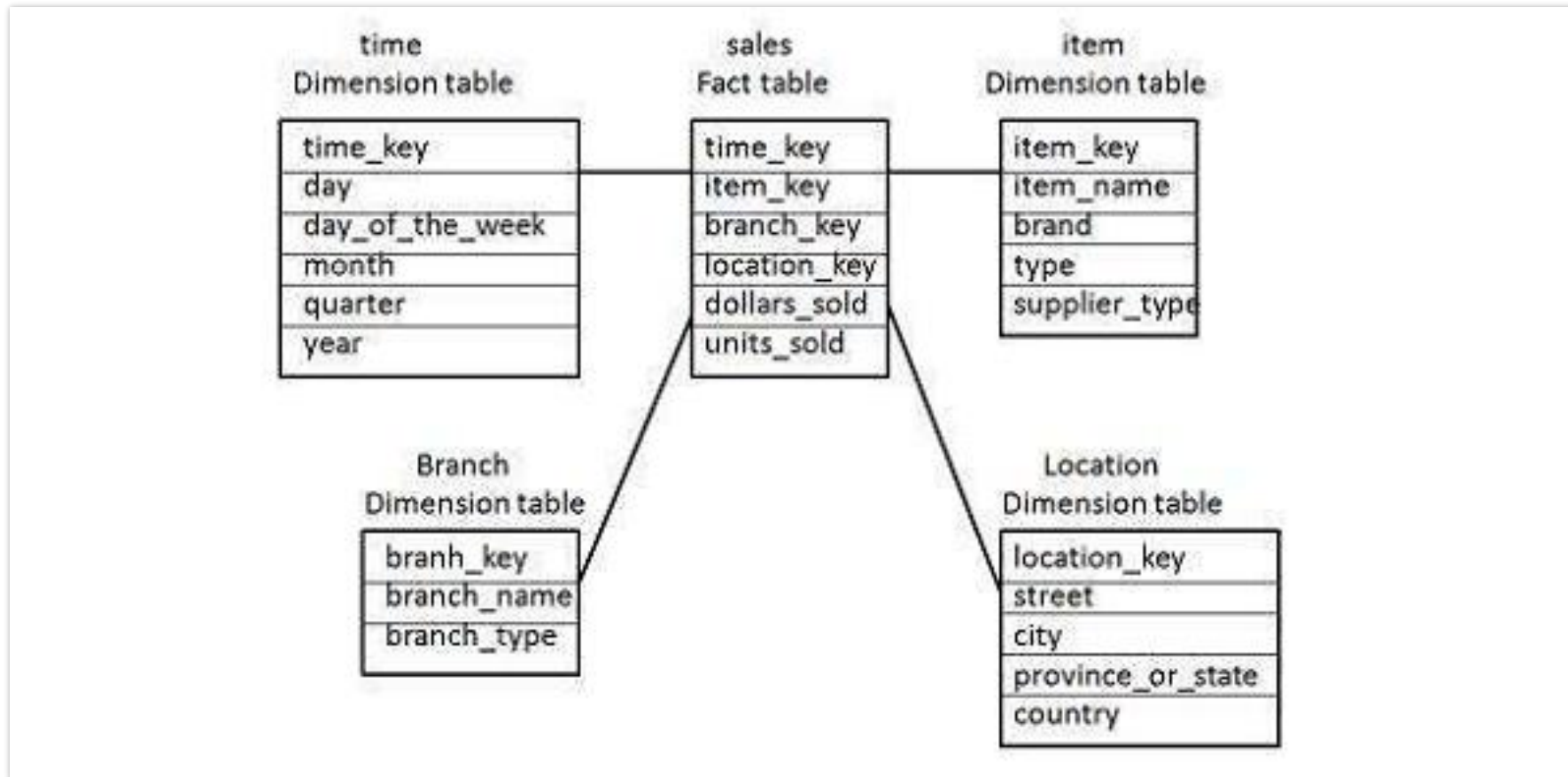
- The cells of the logical cube collected from business operations. 비즈니스 운영으로 얻어진 데이터로 이루어진 논리적 큐브의 각각의 셀들
- Organized by dimensions, which typically include a Time dimension. 차원에 의해 조직 (특히 시간)
- For example, product sales. 예를 들어, 세일 데이터

# Dimensions 차원

- Contain a set of unique values that identify and categorize data. 데이터를 구분하기 위해 사용되는 유일한 값들의 조합
- For example, the sales measure has four dimensions: time, customer, product, and location. 예를 들어, 세일을 구성하는 차원들은 시간, 고객, 상품, 지역

# Star Schema 스타 스키마

- Contains a fact table and multiple dimension tables.  
하나의 팩트테이블과 여러 개의 차원 테이블들로 구성



# 팩트 테이블 Fact Table

- Represents the measures on which analysis is performed. It also contains foreign keys for the dimension keys. 분석할 측정치를 포함. 다른 차원에 대한 외래키를 가짐
- Contain factual or quantitative data (measurements that are numerical, continuously valued, and additive) about a business, such as units sold, orders booked, and so on. 판매 단위, 예약 주문 등과 같은 비즈니스에 대한 실제적 또는 양적 데이터. 숫자, 연속 평가 및 추가되는 측정치가 포함
- Example – every sale is a fact. 예를 들어, 각각의 세일이 팩트임

| Cust Id | Prod Id | Time Id | Qty Sold |
|---------|---------|---------|----------|
| 1110    | 25      | 2       | 125      |
| 1210    | 28      | 4       | 252      |

# Dimension Table 차원 테이블

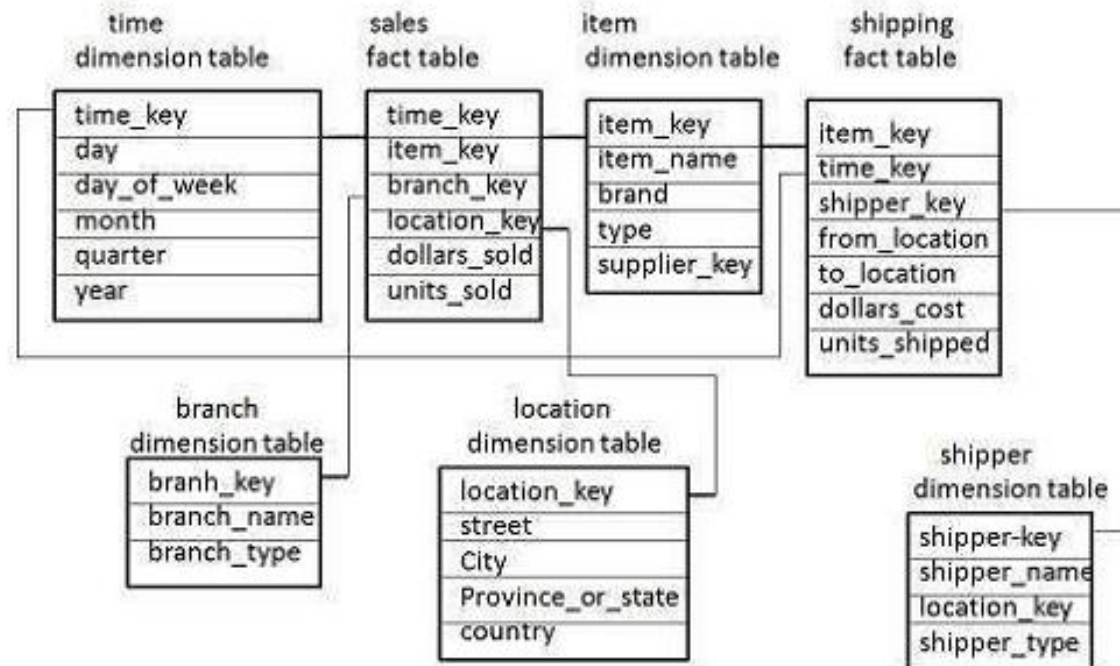
- Represents the characteristics of a dimension. A Customer dimension can have Customer\_Name, Phone\_No, Sex, etc. 각 차원의 특색을 보여줌.  
고객차원의 경우 고객이름, 전화번호, 성별 등
- Usually the source of attributes used to qualify, categorize, or summarize facts in queries, reports, or graphs; thus, dimension data are usually textual and discrete. 일반적으로 쿼리, 보고서 또는 그래프에서 사실을 검증, 분류 또는 요약하는 데 사용되는 속성의 원천. 차원 데이터는 일반적으로 텍스트 및 이산형

| Cust Id | Cust_Name | Phone      | Sex |
|---------|-----------|------------|-----|
| 1110    | Sally     | 1113334444 | F   |
| 1210    | Adam      | 2225556666 | M   |

# Fact Constellation Schema

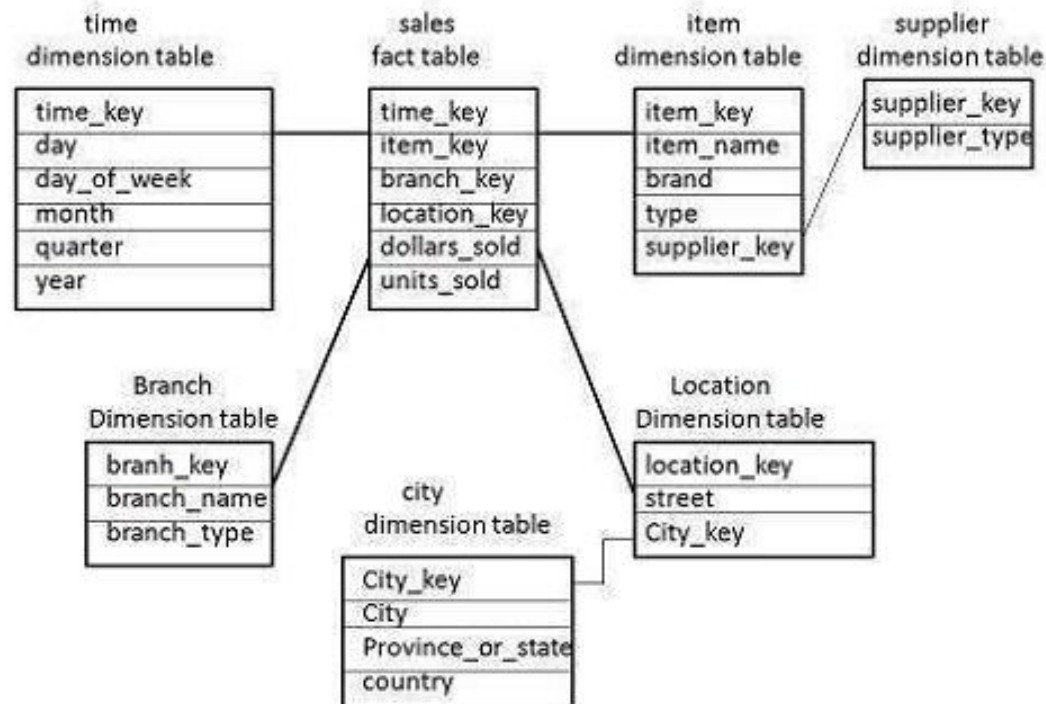
## 팩트 별자리 스키마

- Galaxy Schema. 은하수 스키마
- Has multiple fact tables. 팩트 테이블이 여러 개임

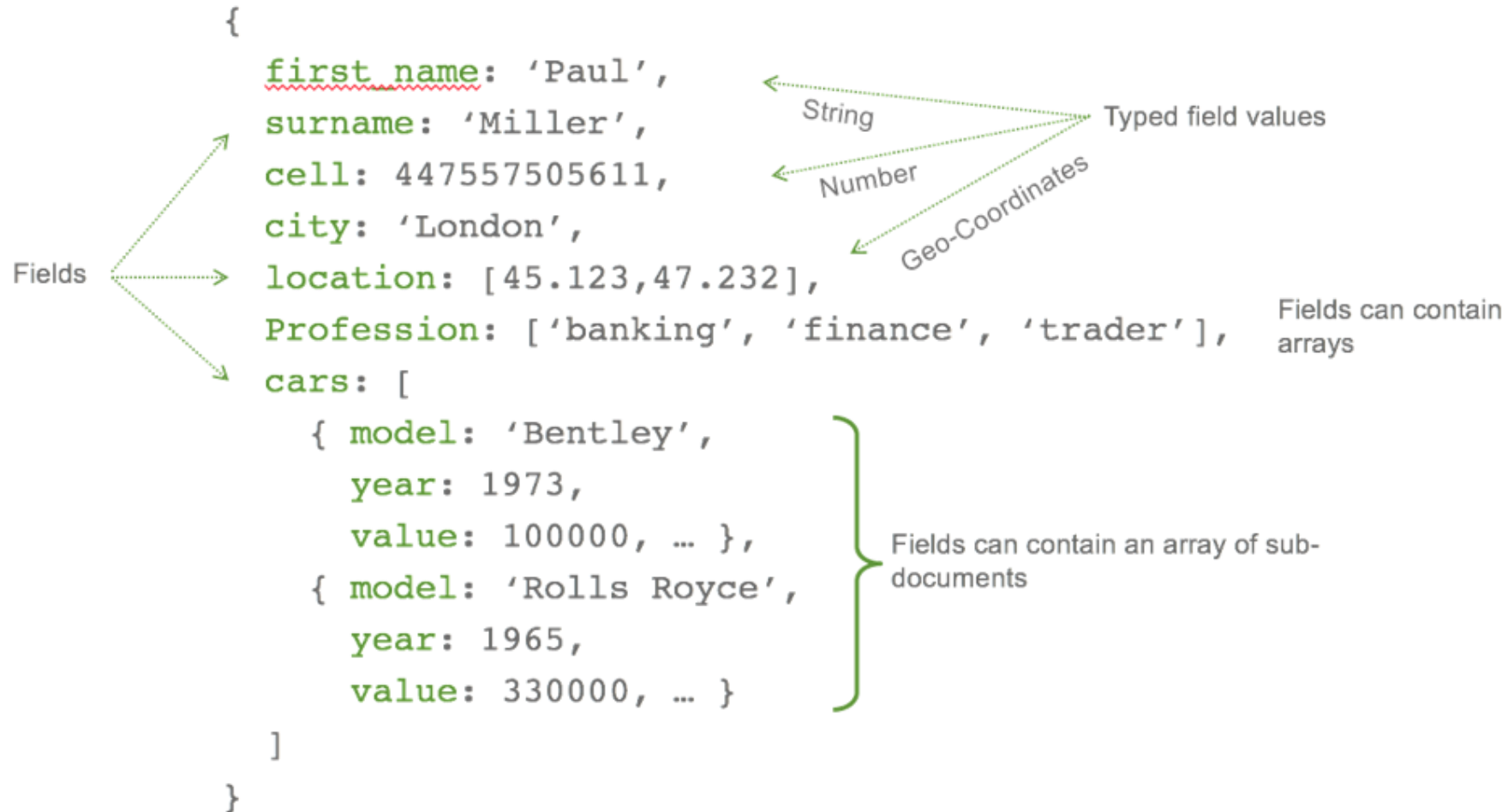


# Snowflakes Schema 눈송이 스키마

- Some dimension tables in the Snowflake schema are normalized. 차원 테이블들이 다시 정규화됨



# MongoDB



# NoSQL Database

- A database that does not use SQL. SQL을 이용하지 않는 데이터베이스
- It is a non-relational database. 관계형 데이터베이스가 아님
- Facebook and Twitter (Web 2.0 Applications) use Apache Software Foundation Cassandra database to quickly create and store massive amounts of data. 페이스북, 트위터는 캐산드라 데이터베이스를 이용하여 많은 양의 데이터를 생성하고 저장

# RDBMS vs MongoDB

| RDBMS       | MongoDB                                                     |
|-------------|-------------------------------------------------------------|
| Database    | Database                                                    |
| Table       | Collection                                                  |
| Tuple/Row   | Document                                                    |
| column      | Field                                                       |
| Table Join  | Embedded Documents                                          |
| Primary Key | Primary Key<br>(Default key _id provided by mongodb itself) |



# SQL vs. NoSQL

## NoSQL



Gaming



Social



IoT



Web



Mobile



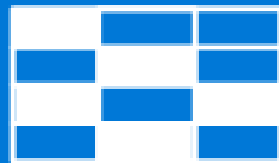
Enterprise



Key/value store



Document  
database



Column family store

## SQL



Web



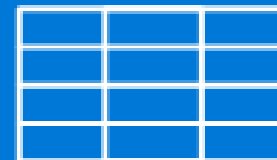
Mobile



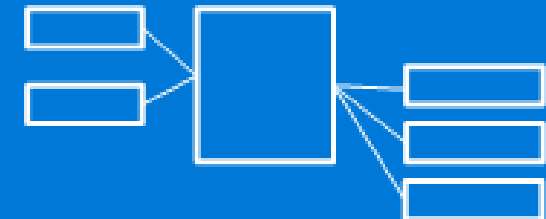
Enterprise



Data mart



Relational table storage



Relationships use joins